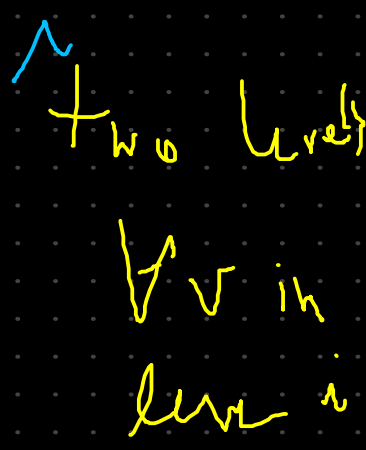
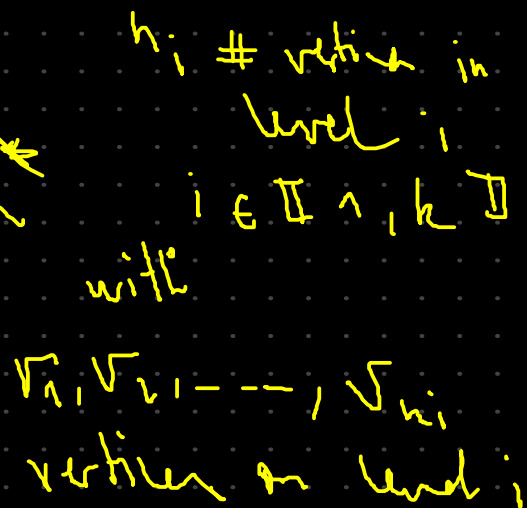
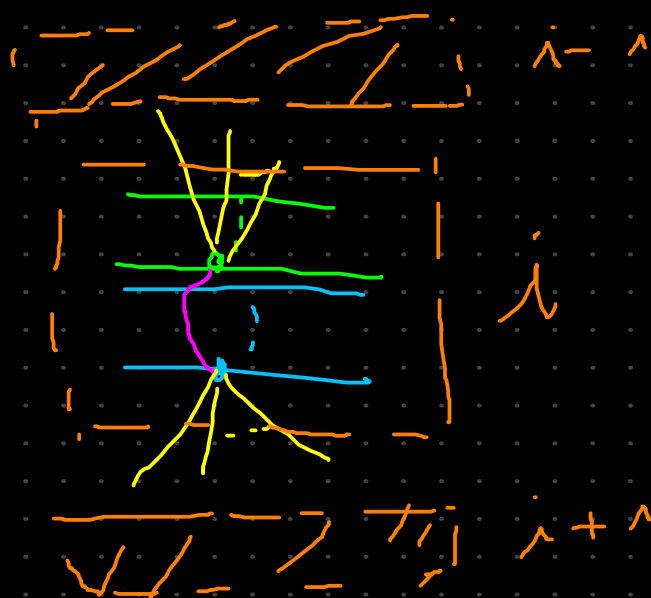


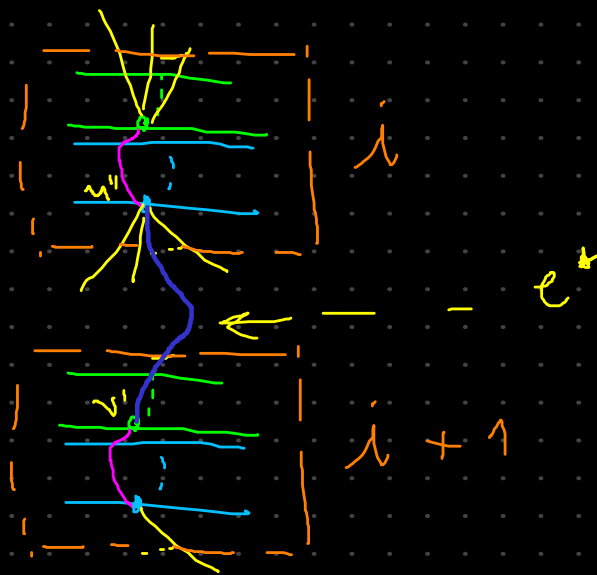
$$V = \left\{ uv \mid u, v \in V, u \neq v \right. \\ \left. \text{and } l(u) = l(v) \right\}$$

$$\left\{ \begin{array}{l} \forall uv \in V : uv \Rightarrow v \sim u \\ \forall uv, wv \in V : uv \wedge wv \Rightarrow uw \\ \forall uv, v'v \text{ with } (u, v), (v, v') \in E \text{ independent} \\ uv \Rightarrow v'v \end{array} \right.$$

proper
level
group
 h^+

one
vertex
per level
graph
 h^*


$$\ell(v_j') = j^i, \ell(v_j'') = n_i + j^i$$




$S'(G)$ 2-SAT $\left\{ \begin{array}{l} \text{independent edges} \\ \text{anti-symmetry} \\ \text{transitivity} \end{array} \right\} S(G)$

Claim: (Random)

$S'(G)$ satisfiable $\Leftrightarrow S(G)$ satisfiable

Claim: (2-SAT & HT-drawings)

$S(G)$ is satisfiable $\Leftrightarrow G$ has a Hamiltonian-Tutte level drawing

HT-drawing

not necessarily
satisfies
planarity

$S(G)$
Random

planar
constraints
not the
transitivity
constraint

a drawing Γ of G

induces

a truth assignment φ of $\mathcal{V}$

$\forall uv \in \mathcal{V}$:
 $\varphi|_{uv} = \text{true}$

$\Leftrightarrow u <_I w$

splitter

$\forall uv \in \mathcal{V}$

$uv \Leftrightarrow \neg wu$

not necessary

$\forall uv, vw \in \mathcal{V} \quad uv \Leftrightarrow vw \quad / \quad (u,v), (w,x) \in E$
independent

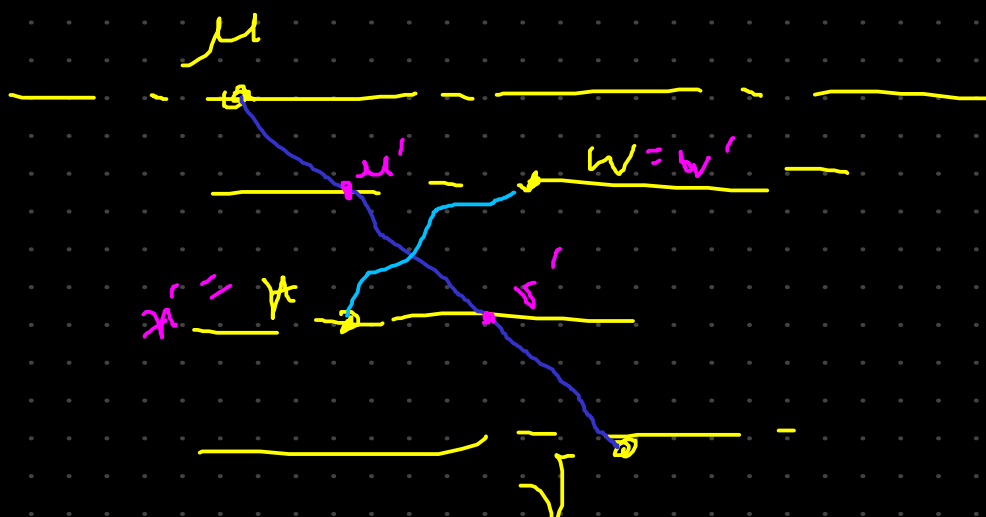
Lemma 2:

(u,v)

initial

(w,x)

$\vdash \Gamma^*(u^*)$



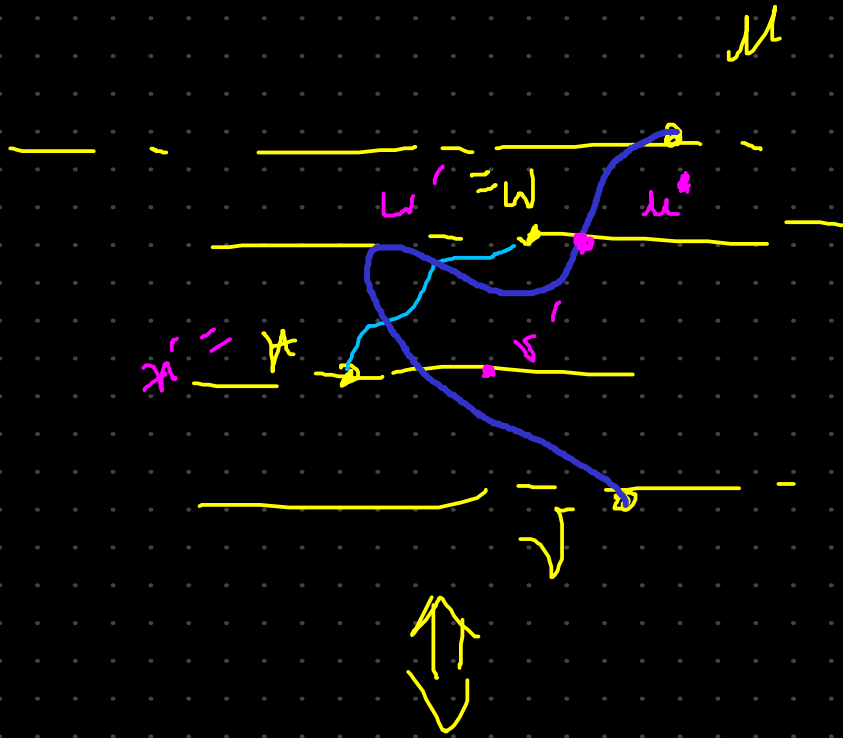
$\triangleright \Gamma^+(u^*)$

induced

from $\Gamma^*(u^*)$

let φ^+ be the truth assignment of $S(w^+)$ induced by Γ^+

$\Rightarrow$



$$\varphi^+(u'w') = \varphi^+(v'x')$$

done.

stopped in lemma 2. I will need to pick up on it tomorrow